

Research Readiness Statement

a. Degree program and mentor preparation. Colton Goodrich is a PhD student in Geosensing Systems Engineering & Sciences at the University of Houston, advised by the proposal PI, Craig L. Glennie. Graduate coursework in remote sensing and geosensing, together with prior degrees in Geology (M.S. and B.S., Brigham Young University) and a Geospatial Analysis certificate (Weber State University), provide a foundational understanding of terrain analysis, digital elevation models, and remote sensing relevant to the proposed research. The proposed project requires use of the Python scientific stack (NumPy, GDAL, Rasterio, PyTorch), PDAL, WhiteboxTools, and QGIS, together with DEM differencing, co-registration (ICP), and robust statistics; the FI is already proficient in each of these tools and methods. Two areas require further preparation — hierarchical/Bayesian statistical modeling and cold-regions (Antarctic) process geomorphology; the FI will address these deficiencies through graduate coursework and directed study with the PI.

b. Graduate study timeline. Degree type: PhD. Subject area: Geosensing Systems Engineering & Sciences. Enrollment: enrolled since August 2025 (approximately one year at the proposal due date). Estimated graduation: August 2029.

c. Other relevant experience. The FI independently designed and built an end-to-end airborne-lidar terrain-analysis pipeline on the Appalachian Plateau — point-cloud processing (PDAL), bare-earth derivative generation (GDAL/WhiteboxTools), hand-labeled training-set construction (QGIS), U-Net and instance-segmentation model training (PyTorch), and DEM-of-Difference change analysis with ICP co-registration and NMAD-based uncertainty — an independent research project that exercises the methods used in the proposed work. Additional relevant experience includes InSAR-based DEM generation and accuracy assessment against an airborne lidar reference; five years as an Environmental Scientist at the Utah Division of Oil, Gas & Mining (geospatial data analysis and Python automation); prior work as an Earth Scientist at Chevron; a graduate teaching assistantship at Brigham Young University; and a Python Programming course (The Tech Academy, 2024).

NASA BIOGRAPHICAL SKETCH FORM

Identifying Information

Name: Goodrich, Colton, Lynn

Persistent Identifier (PID): ORCID 0009-0006-5237-2836 (<https://orcid.org/0009-0006-5237-2836>)

Position Title: Doctoral Researcher, Geosensing Systems Engineering & Sciences

Organization and Location: University of Houston, Houston, TX, USA

Professional Preparation

University of Houston — Houston, TX, USA

Degree: Ph.D., Geosensing Systems Engineering & Sciences. Start: 08/2025. Received: Expected 08/2029. Field: Geosensing / Remote Sensing

Weber State University — Ogden, UT, USA

Degree: Certificate, Geospatial Analysis. Start: 2018. Received: 2019. Field: Geospatial Analysis / GIS

Brigham Young University — Provo, UT, USA

Degree: M.S. Start: 09/2011. Received: 12/2013. Field: Geology

Brigham Young University — Provo, UT, USA

Degree: B.S. Start: 2006. Received: 2011. Field: Geology

Appointments and Positions

2025–Present · Doctoral Researcher

University of Houston, Geosensing Systems Engineering & Sciences · Houston, TX, USA

2020–2025 · Environmental Scientist III (promoted from Environmental Scientist II)

Utah Department of Natural Resources, Division of Oil, Gas & Mining · Syracuse, UT, USA

2013–2016 · Earth Scientist

Chevron Energy Company · Houston, TX, USA

2012–2012 · Reservoir Modeling Intern

Chevron Energy Company · Houston, TX, USA

2011–2013 · Graduate Teaching Assistant

Brigham Young University, Department of Geology · Provo, UT, USA

Products

1. Goodrich, C. (2025). WellSight: An autonomous lidar analysis pipeline for detecting orphaned and abandoned oil & gas wells [software]. Multi-task deep-learning system (U-Net, ~8M parameters) segmenting well pits, access roads, and well pads from lidar-derived rasters at 0.5–1 m resolution; integrates PDAL, WhiteboxTools, GDAL/Rasterio, and PyTorch.
https://github.com/clgoodrich/lidar_project

2. Goodrich, C. (2025). SAOCOM L-band InSAR digital elevation model generation and validation framework [software/dataset]. P-SBAS time-series interferometry of SAOCOM-1A/1B data producing high-coherence, vegetation-penetrating DEMs; accuracy assessment (NMAD, RMSE, CORINE land-cover-stratified) against TINITALY lidar and Copernicus GLO-30, in support of the NASA Surface Topography and Vegetation (STV) Decadal Survey. https://github.com/clgoodrich/saocom_project
3. Goodrich, C. (2025). Coastal land-subsidence mapping from Small Baseline Subset (SBAS) InSAR time series [software]. Deformation-rate estimation on HyP3 SAR processing outputs with an accuracy-validation workflow. https://github.com/clgoodrich/Final_project_coastal
4. Goodrich, C. (2024). EToolsV3 — Regulatory GIS web application for directional-survey trajectory computation and Well Completion Report generation [software]. Minimum-curvature trajectory recomputation, PLSS location, boundary-footage calculation, LLM-assisted PDF extraction, interactive 2D/3D plats (FastAPI, NiceGUI, Shapely/GeoPandas). <https://github.com/clgoodrich/EToolsV3>
5. Goodrich, C. (2024). ArcGIS geoprocessing toolbox for social-media geospatial analysis [software]. Parameterized ArcPy tools (kernel-density hotspots, H3 hexagon aggregation, time-enabled mosaics). https://github.com/clgoodrich/tweet_project_v2
6. Goodrich, C. (2024). 1 m DEM/hillshade mosaic over Oil Creek State Park, Pennsylvania [dataset]. Contiguous 22-tile (~7.5 × 9 km) mosaic assembled via a greedy contiguous-patch downloader over the USGS TNM Access API. https://github.com/clgoodrich/lidar_project
7. Goodrich, C. (2024). OGM Board regulatory mapping and visualization application [software]. Decision-support application presenting well data, analyses, and digital maps for monthly hearings of the Utah Board of Oil, Gas & Mining. <https://github.com/clgoodrich/OGMBoardProject>

Certification

I certify that the information provided is current, accurate, and complete. This includes but is not limited to information related to domestic and foreign appointments and positions.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§ 287, 1001, 1031 and 31 U.S.C. §§ 3729-3733 and 3802.

Signature: Colton Goodrich Date: 07/14/2026

*Privacy Act and Burden Statement:

The information requested on this form is solicited under the authority of the National Aeronautics and Space Act of 1958 (51 U.S.C. 20101 et. seq.) The information on this form will be used in connection with the evaluation of qualified proposals. The information requested may be disclosed to qualified reviewers for their opinion and evaluation of applicants and their proposals as part of the NASA application review process; to other government agencies or other entities needing information regarding applicants or nominees as part of a joint application review process, or in order to coordinate programs or policy; or to individual or institutional applicants and award recipient institutions to provide or obtain data as part of the application review process, award decisions, or administering awards. Additionally, information requested may be disclosed to other entities when merging records with other computer files to carry out studies for or otherwise assist NASA with program management, evaluation, or reporting as well as contractors,

grantees, volunteers, experts, consultants, advisors, and other individuals who perform a service to or work on or under a contract, grant, cooperative agreement, advisory committee, independent review boards, or other arrangement with or for NASA or for the Federal government. See NASA Systems of Records Notice (SORN) NASA 100AAR, "Opportunities and Associated Reviewers."

This information collection meets the requirements of 44 U.S.C. § 3507 as amended by section 2 of the Paperwork Reduction Act of 1995. You do not need to answer these questions unless we display a valid Office of Management and Budget (OMB) control number. The OMB control number for this collection is 3145-0279 and this information collection expires on October 31, 2026. We estimate that it will take two (2) hours to read the instructions, gather the facts, and answer the questions. Send only comments relating to our time estimate to: christiane.diallo@nasa.gov.

NASA CURRENT AND PENDING (OTHER) SUPPORT FORM

Identifying Information

Name: Goodrich, Colton, Lynn

Persistent Identifier (PID): ORCID 0009-0006-5237-2836

Position Title: Doctoral Candidate, Geosensing Systems Engineering & Sciences

Organization and Location: University of Houston, Houston, TX, USA

a. Proposals and Active Projects

Entry 1 — Graduate Assistantship

Title: Graduate assistantship — monthly stipend and tuition waiver

Status of Support: Current

Award Number (if available):

Source of Support: University of Houston, Cullen College of Engineering, NCALM

Primary Place of Performance: University of Houston, Houston, TX, USA

Start Date: 08/2025 **Expected End Date:** 08/2029

Total Anticipated Amount: Stipend \$2,601.67/month (~\$31,220/year, 12-month appointment) + tuition waiver ~\$3,852/semester (~\$7,700/year; TX-resident Engineering-PhD rate, 9 SCH, AY2027; mandatory fees additional)

Person-Months Per Year: 12

Overall Objectives: Graduate assistantship supporting the FI's doctoral study, training, and dissertation research in geosensing/remote sensing; not tied to a single external project.

Statement of Potential Overlap: No scientific overlap with the pending FINESST proposal (Entry 2). If FINESST is awarded, 6 person-months/year now covered by this assistantship would be reallocated to FINESST for the proposed research; the assistantship adjusts accordingly, so the FI's total committed effort stays at 12 person-months/year with no double compensation.

Entry 2 — This FINESST proposal

Title: Detection to Attribution: Linking Two Decades of Antarctic Stream-Channel Change to Climate Drivers Utilizing Terrain Data

Status of Support: Pending

Award Number (if available): N/A (under review)

Source of Support: National Aeronautics and Space Administration (NASA) — FINESST, ROSES-2025 Appendix F.5

Primary Place of Performance: University of Houston, Houston, TX, USA

Start Date: 01/2027 **End Date:** 08/2029 (2 years, 8 months)

Total Anticipated Amount: up to \$133,000 (\$50,000 per budget year, final year partial)

Person-Months Per Year: 6

Overall Objectives: Attribute two decades of measured stream-channel and stream-corridor geomorphic change in the McMurdo Dry Valleys to climate drivers, using multi-epoch lidar and satellite-DEM differencing with per-pixel calibrated uncertainty.

Statement of Potential Overlap: This is the proposal under consideration. No scientific overlap with current support; any budgetary/effort overlap with the assistantship is resolved by reallocation so total committed effort remains ≤ 12 person-months/year.

Certification

I certify that the information provided is current, accurate, and complete. This includes, but is not limited to, information related to current, pending, and other support (both foreign and domestic) as defined in 42 U.S.C. §6605.

I also certify that, at the time of submission, I am not a party to a malign foreign talent recruitment program.

Misrepresentations and/or omissions may be subject to prosecution and liability pursuant to, but not limited to, 18 U.S.C. §§287, 1001, 1031 and 31 U.S.C. §§3729-3733 and 3802.

Signature: Colton Goodrich Date: 7/14/2026

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FINESST Ancillary Documents

A. Facilities, Equipment, and Resources Statement

The proposed work requires modest, presently available resources. There is no need for facility development or shared instrument time. The FI has easy access to all of the resources listed below, so no resource-support letter is required.

Computing. The FI has a workstation with a single consumer-grade GPU (NVIDIA GeForce GTX 1070 Ti, 8 GB of VRAM) and at least 2 TB of local storage, owned by the research group and the FI. This is more than sufficient for all model training and elevation processing in the proposal.

Workspace. The FI has office and lab space in NCALM (National Center for Airborne Laser Mapping) at the University of Houston, with institutional backup storage for project data.

Software. The full processing stack is open source and requires no subscriptions or commercial licenses (Python scientific stack, PDAL, GDAL/OGR, WhiteboxTools, PyTorch, QGIS).

Field work. None is proposed. The project is based entirely on existing datasets and requires no Antarctic deployment, logistics support, or permitting.

B. Acknowledgements

The Future Investigator (FI) is the primary author of this proposal. The FI thanks Dr. Craig Glennie for his scientific mentorship and guidance in shaping and editing the research, and Cami Barlow, whose dissertation on Antarctic stream-channel detection and geomorphic change analysis provides the foundation this project builds upon. AI assistance (Anthropic's Claude, used as an editing tool under the FI's direction) contributed to light grammar and structure editing, document formatting, and data-availability verification; all scientific content, claims, and decisions are the FI's own and were reviewed by the FI and PI. Publicly archived datasets are credited to their providers: NASA ATM (via USGS), NCALM/OpenTopography, the Polar Geospatial Center (REMA), the McMurdo Dry Valleys LTER (via the Environmental Data Initiative), Copernicus/ECMWF (ERA5), and NCAR (AMPS).

C. Budget Justification / Narrative Skeleton

Item	Amount	Justification
FI stipend		12-month graduate research assistantship at \$2,601.67/month; the FI performs all proposed research
Tuition & fees		Consolidated tuition, 2 semesters $\times$ \$3,852 (TX-resident Engineering-PhD rate, 9 SCH, AY2027); add mandatory fees
Conference travel		AGU (Fall Meeting), cryosphere section — poster in Yr 1, talk in later years
Publication		Open-access fee, one first-

		author paper
Computing/storage		Local storage expansion for full-valley raster stacks; no cloud compute required
Year 1 total (direct)		Under the \$50,000/year cap

- No PI salary (FINESST does not fund the PI); no field-work or logistics costs.
- F&A/indirect: not permitted for scholarship awards; if the FI is treated as an employee (GRA), confirm allowable rates with the research office so the total stays under the \$50,000/year cap.
- Award total (01/2027 to 08/2029, ending at the FI's graduation) stays under the \$150,000 cap after stipend/tuition escalation.